Martin DAVILA

Software Engineer · Full-Stack Development · Bioinformatics · AI/ML

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PROFESSIONAL SUMMARY

Software engineer with 5+ years of experience building production systems in healthcare, genomics, and SaaS, working across full-stack development, bioinformatics, and AI/ML. I work mainly in Python, TypeScript, and C#, with React, Next.js, ASP.NET Core, and FastAPI, and I am just as comfortable setting up AWS and Azure infrastructure and CI/CD pipelines, or building ML models from scratch. My scientific background helps me turn complex biological data problems into software that scales and stays easy to maintain. I have worked in regulated environments that handle sensitive data, and I collaborate closely with scientists, clinicians, and engineering teams.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C#, PHP 8, R, SQL, Kotlin

Frontend: React, Next.js, Angular, Tailwind CSS, Material UI, HTML5, CSS3, Sass

Backend / APIs: Node.js, NestJS, FastAPI, Flask, Express.js, ASP.NET Core, REST APIs, microservices

Cloud / DevOps: AWS (EC2, S3, Lambda, RDS), Azure, Docker, Kubernetes, Jenkins, GitLab CI/CD, GitHub Actions, Terraform

Data / AI / ML: Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, SciPy, NLP, predictive modeling

Bioinformatics: ChIP-Seq, RIP-Seq, Hi-C, Biopython, BLAST, Bowtie, Cufflinks, genomic pipelines, LIMS

Tooling: TDD, Jest, Pytest, Selenium, Git, GitLab, Jira, Agile/Scrum, Cursor, Claude Code, GitHub Copilot

WORK EXPERIENCE

Eurofins Biomnis · Lyon, France — Aug 2024 – Present

Senior Software Engineer — Full-Stack & Bioinformatics

- Architected and shipped the Prescriber Portal — a regulated clinical web application (C# / ASP.NET Core + Angular) enabling 3,000+ medical prescribers across France to securely access laboratory diagnostics and patient reports.
- Built Python bioinformatics automation tools that cut manual genomic analysis time by ~65%, freeing lab analysts for higher-value diagnostic work.
- Rolled out CI/CD pipelines and TDD across the engineering team; raised test coverage from near zero to 92% and reduced code-to-production lead time from 5 days to under 24 hours.
- Integrated AI-assisted development tooling (Cursor, GitHub Copilot) into a regulated healthcare workflow without compromising audit trail or compliance requirements.

Agilebio · Paris, France (Remote) — Mar 2024 – Jul 2024

DevOps Engineer & Full-Stack Developer

- Modernized CI/CD pipelines for LabCollector LIMS (Jenkins + AWS), cutting deployment time by 40% and enabling 3× more frequent releases with no regression increase.
- Refactored core PHP 8.2 application code; improved response times by ~22% on high-traffic diagnostic endpoints through profiling and targeted optimization.
- Led containerization groundwork using Docker and Kubernetes, reducing per-environment provisioning from several hours to under 15 minutes.

iGEM — International Genetically Engineered Machine · Paris, France (Remote) — Feb 2023 – Oct 2023

Software Engineer — Front-End & Back-End

- Built React 18 / TypeScript UI features for the global genomics platform; improved Lighthouse performance score from 62 to 84 through targeted Core Web Vitals optimizations.
- Developed NestJS REST APIs and AWS integrations handling genomic submission data for 6,000+ teams across 40+ countries.
- Established structured code review standards across the team; reduced average bug-introduction rate per sprint by ~30%.

Hiring for Good · Paris, France (Remote) Mar 2023 – Oct 2023

Full-Stack R&D Engineer

- Led end-to-end development of Profiler, an AI-powered HR SaaS platform (Next.js + Python Flask + AWS), from zero to MVP in 14 weeks — delivered two weeks ahead of schedule.
- Built ML candidate-matching pipeline using Scikit-learn (Logistic Regression, Random Forest, SVM); achieved 82% precision in production, reducing average time-to-shortlist by 3 weeks.

INSA Lyon · Lyon, France Jan 2019 – Jan 2022

Doctoral Researcher — Bioinformatics

- Conducted research on 3D chromosomal organization in Dickeya dadantii during host infection; contributed to 2 peer-reviewed publications and 3 international conference presentations.
- Built Python automation pipelines for ChIP-Seq, RIP-Seq, and Hi-C genomic data processing, reducing per-experiment analysis time from ~8 hours to under 2 hours.

University of Valladolid · Valladolid, Spain Jan 2018 – Jan 2019

Research Assistant — Bioinformatics

- Developed a Python bioinformatics pipeline for genetic deletion analysis, cutting researcher turnaround time by ~40% and enabling cross-institutional data sharing across 3 partner labs.

SELECTED PROJECTS

Prescriber Portal — Eurofins Biomnis · prescriberportal.eurofins-biomnis.com

C#, ASP.NET Core, Angular, OIDC, healthcare data-protection compliance

Production clinical portal enabling 3,000+ medical prescribers to securely access laboratory diagnostics across the Eurofins Biomnis network in France.

imartin.dev — Personal Portfolio · imartin.dev/

Next.js 15, React 19, TypeScript, Tailwind CSS, React Hook Form, Zod, Vercel

Full-stack portfolio with analytics integration, achieving a 98/100 Lighthouse performance score.

Predictive Employee Retention Model

Python, Scikit-learn, Logistic Regression, Random Forest, SVM

ML ensemble model achieving 82% churn-prediction precision; contributed to a measured 15% reduction in employee turnover for Hiring for Good clients.

EDUCATION

INSA Lyon · Lyon, France 2019 – 2022

Doctoral Research in Bioinformatics

University of Valladolid · Valladolid, Spain 2018 – 2019

MSc in Chemical Engineering & Environmental Technology · Engineering Degree in Biotechnology

LANGUAGES

English: Professional | **French:** Professional